A

Report on

PRACTICAL SECURITY ASSESSMENT (TASK 8)

Submitted In Partial Fulfillment of the Requirements

for the

Cyber Security Internship (Week 3) (VAPT)

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CyArt Tech LLP

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1. INTRODUCTION

1.1 Overview

Vulnerability Assessment and Penetration Testing (VAPT) is a critical cybersecurity process used to identify, analyze, and mitigate security vulnerabilities in systems, networks, and applications. It combines automated scanning with manual testing to simulate real-world attacks.

1.2 Purpose of This Task

The purpose of this report is to understand advanced exploitation techniques, web application testing, and effective reporting practices. It aims to build practical and theoretical knowledge required for real-world cybersecurity roles.

1.3 Target Information

- Target Type: Web Application / Virtual Machine
- Environment: Controlled Lab Setup (DVWA, Metasploitable, VulnHub)
- Tools Used: Metasploit, Burp Suite, sqlmap, OWASP ZAP

1.4 Objectives

- Learn advanced exploitation techniques
- Identify and exploit web vulnerabilities
- Understand reporting standards
- Improve communication with stakeholders

2. ADVANCED VULNERABILITY EXPLOITATION

2.1 Exploit Chains

Exploit chains involve combining multiple vulnerabilities to achieve a larger attack goal. For example, an attacker may use Cross-Site Scripting (XSS) to steal session cookies and then perform session hijacking to gain unauthorized access.

Figure 1: Exploit chain diagram

Figure 2: Multistage cyber attack life cycle

2.2 Exploit Customization

Exploit customization refers to modifying existing exploit scripts or payloads to match a specific target environment. This may include changing IP addresses, ports, encoding methods, or payload delivery techniques to ensure successful execution.

Figure 3: Exploit Development Life Cycle

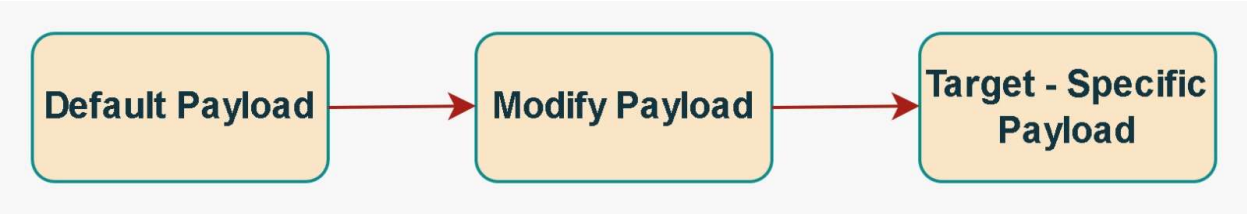

Figure 4: Payload Customisation Process

2.3 Obfuscation Techniques

Obfuscation techniques are used to bypass security mechanisms such as Web Application Firewalls (WAFs). Common methods include:

- Encoding payloads (Base64, URL encoding)
- Using polymorphic code
- Splitting malicious payloads

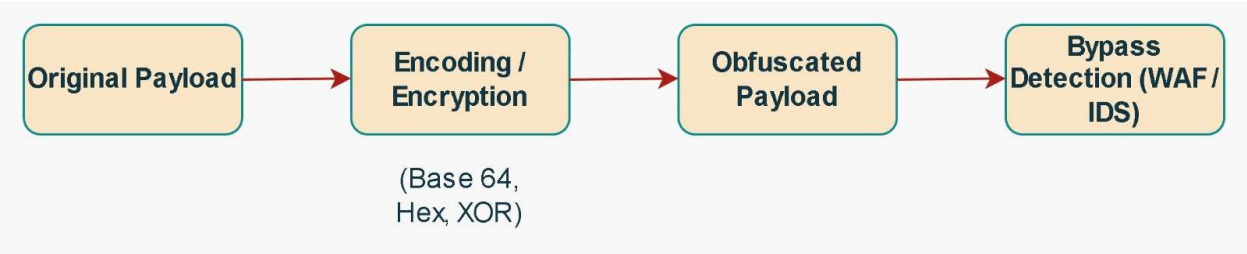

Figure 5: Payload Obfuscation Techniques

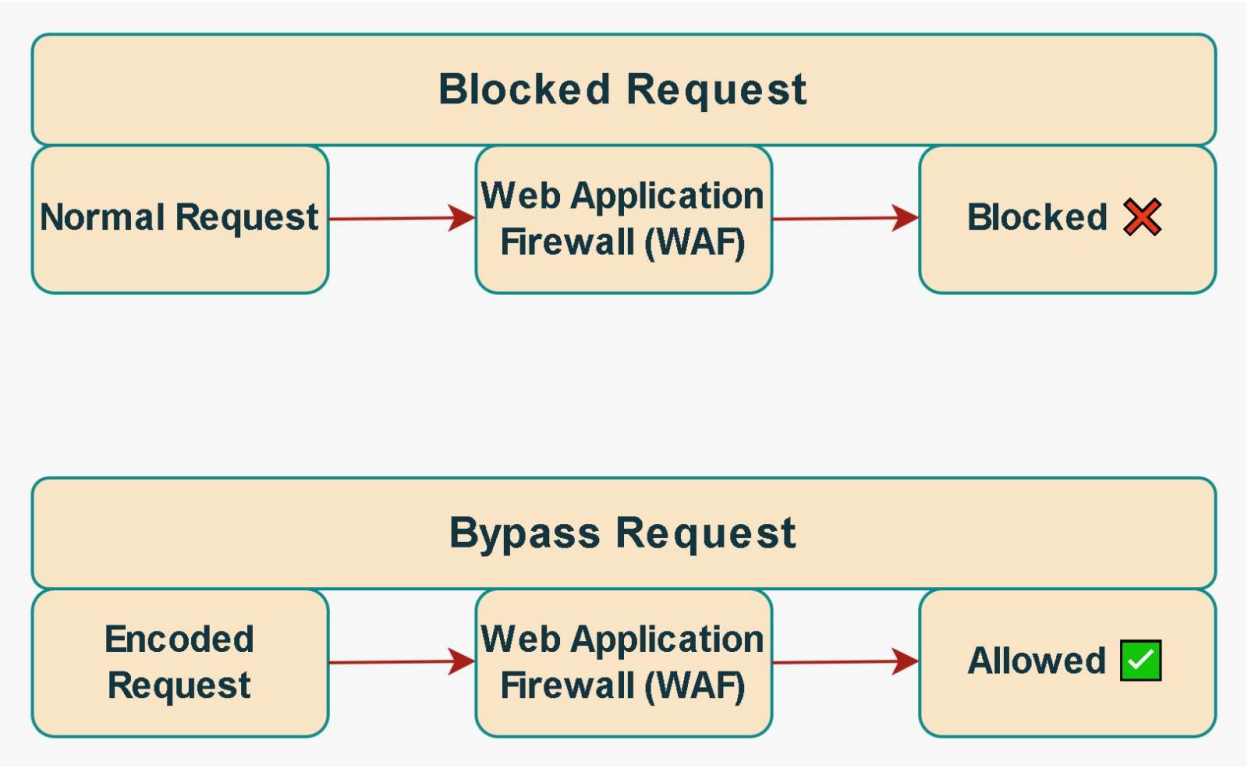

Figure 6: WAF Bypass Techniques using Encoding

3. WEB APPLICATION PENETRATION TESTING

3.1 OWASP Top 10

The OWASP Top 10 is a widely recognized standard for web application security risks.

Common vulnerabilities include:

- Injection Attacks (SQL Injection)
- Broken Authentication
- Security Misconfiguration
- Cross-Site Scripting (XSS)

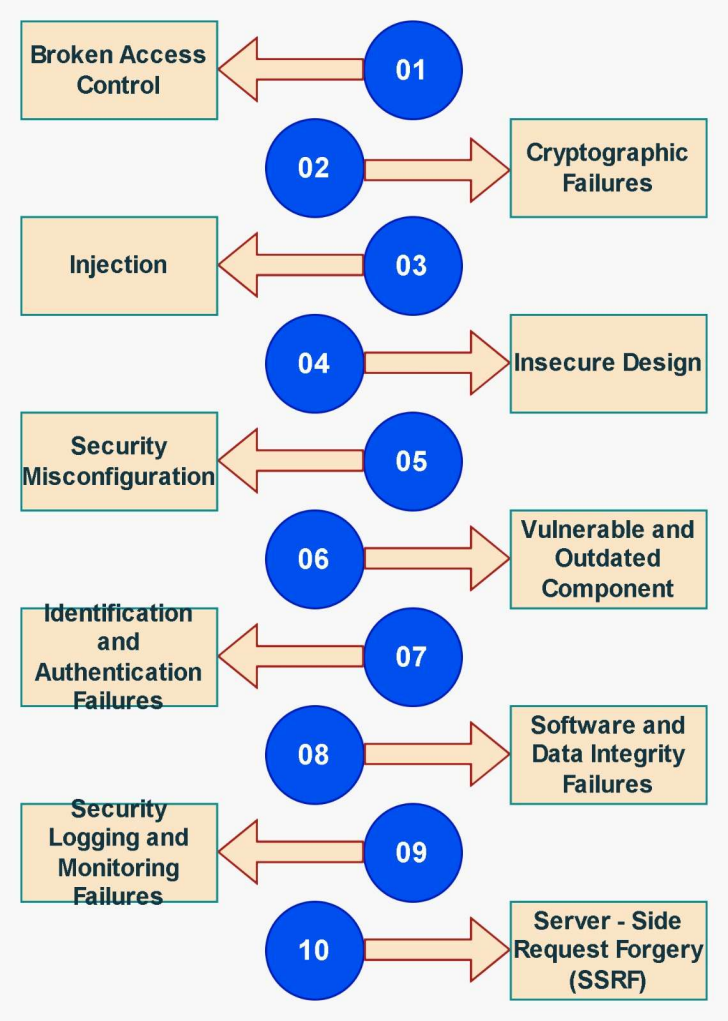

Figure 7: OWASP Top 10 Overview

3.2 Testing Techniques

Web application testing includes:

- Manual Testing: Using tools like Burp Suite to intercept and manipulate requests
- Automated Testing: Using tools like sqlmap and OWASP ZAP
- Fuzzing: Sending unexpected inputs to identify vulnerabilities

Figure 8: Web Application Attack Flow

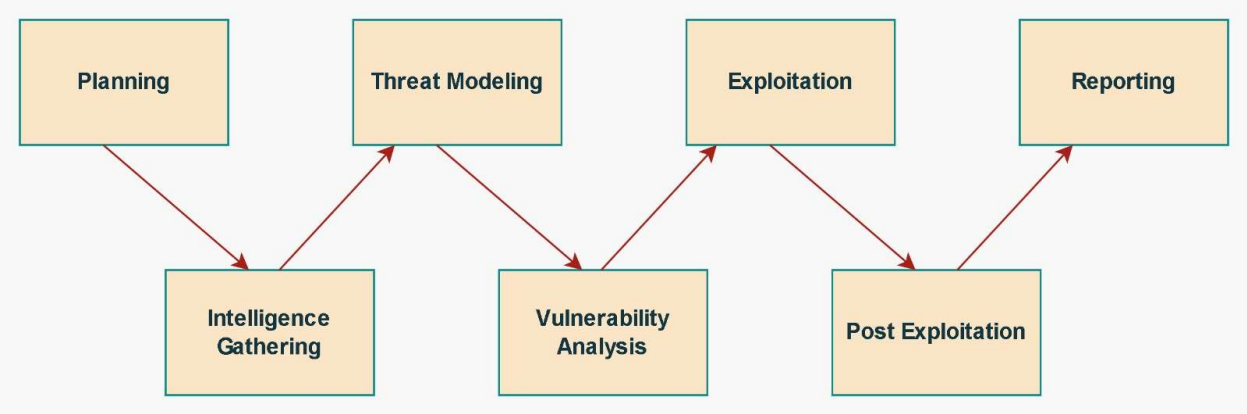

Figure 9: Web Penetration Testing Process

3.3 Secure Coding Practices

Secure coding helps prevent vulnerabilities by:

- Validating user input
- Using parameterized queries
- Implementing strong authentication
- Managing sessions securely

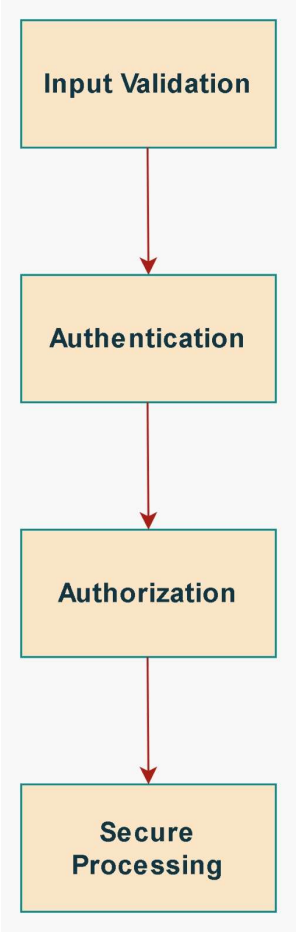

Figure 10: Secure Coding Practices

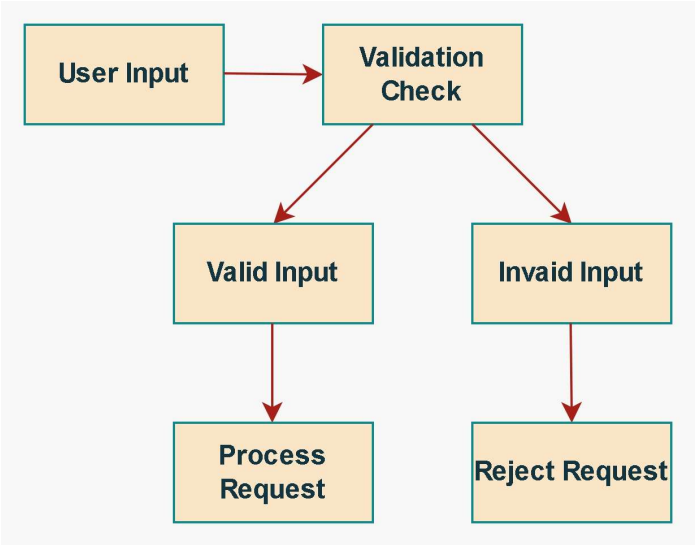

Figure 11: Input Validation Workflow

4. REPORTING & STAKEHOLDER COMMUNICATION

4.1 Report Structure

A standard VAPT report includes:

- Executive Summary
- Technical Findings
- Risk Assessment
- Remediation Steps

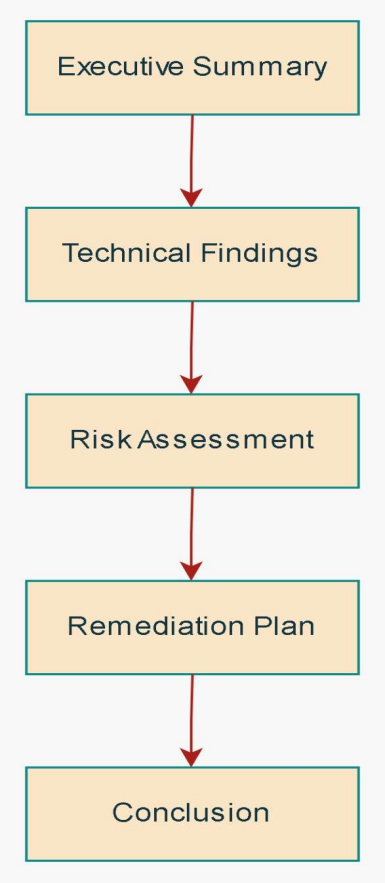

Figure 12: VAPT Report Structure

4.2 Risk Assessment (CVSS)

The Common Vulnerability Scoring System (CVSS) is used to measure the severity of vulnerabilities. It considers:

- Exploitability
- Impact
- Environmental factors

Scores range from Low to Critical severity.

Low	Medium	High	Critical
0.0 - 3.9	4.0 - 6.9	7.0 - 8.9	9.0 - 10

Figure 13: Risk Assessment using cvss

CVSS v 3.0 Ratings			
Low	Medium	High	Critical
0.1 - 3.9	4.0 - 6.9	7.0 - 8.9	9.0 - 10

Figure 14: Vulnerability Severity Classification

4.3 PTES Reporting

Penetration Testing Execution Standard (PTES) provides guidelines for reporting:

- Clear documentation of findings
- Evidence of exploitation
- Step-by-step reproduction
- Remediation recommendations

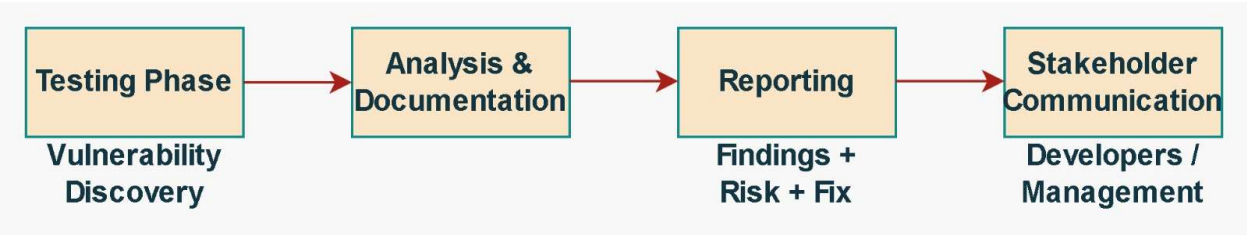

Figure 15: PTES Reporting Workflow

4.4 Stakeholder Communication

Effective communication ensures that:

- Managers understand risks in simple terms
- Developers receive technical details for fixing issues
- Reports are clear, concise, and actionable

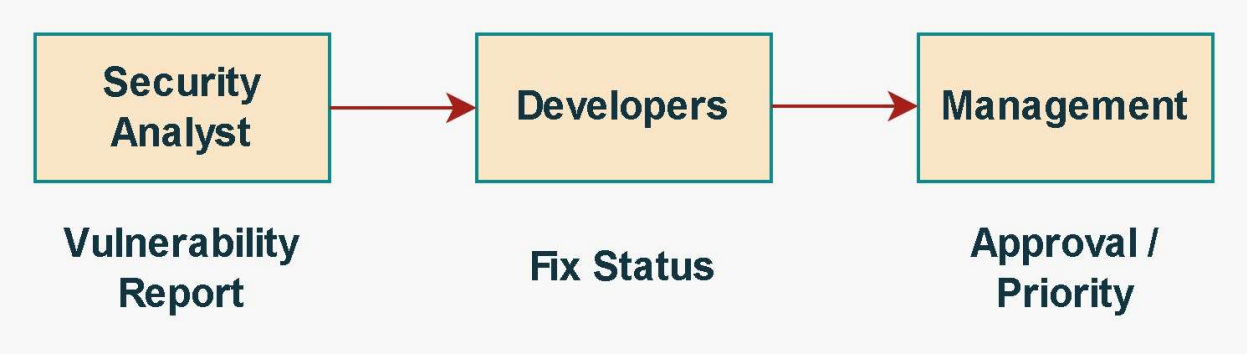

Figure 16: Stakeholder Communication Flow

5. CONCLUSION AND REFERENCE

5.1. Conclusion

This report highlights the importance of advanced vulnerability exploitation, web application penetration testing, and structured reporting. By understanding exploit techniques and applying proper testing methodologies, organizations can identify and mitigate risks effectively. Strong communication and reporting ensure that vulnerabilities are addressed promptly, improving overall security posture.

5.2 References

- OWASP Web Security Testing Guide (WSTG)
- OWASP Top 10 (2021)
- PTES (Penetration Testing Execution Standard)
- Exploit-DB
- PortSwigger Web Security Academy
- SANS Cybersecurity Resources